



# DNA OF THE FUTURE "Genes have very little impact on life outcomes"

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**R**apid changes have taken place in genomics over a short period of time. In 1995, I led the team that for the first time sequenced the genome of a living organism. This was a computationally-intense project. We scaled up very rapidly and in 2001 were able to publish the sequence of the first human genome. Today, we have re-decoded it. In September 2007, we published a complete diploid genome—the newest stage in human genome sequencing—in contrast to half the genomes cracked at the turn of the decade. A diploid genome consists of the DNA in both sets of chromosomes, one from each parent, and it is the normal genome possessed by almost all the body cells. The genome we decoded was based mostly on my own DNA and it included chromosomes from both my parents.

And it totally changed our view. The new genome sequence made it clear that the variation in the genetic programming carried by an individual is much greater than expected. We are 1 to 2 per cent different from each other instead of 0.1 per cent, as thought earlier. That's a huge change. The assumption earlier was that we all had the same human genes and that we just differed from each other in one letter out of a thousand. Now we know that we all don't have the same sets of gene and we differ quite substantially in those genes. Forty-four per cent of my genes had what we call heterozygote variations. The two chromosomes differed from each other in terms of the gene structure. So there's a huge degree of human variation if we all have two different versions of genes. You can imagine that there could be billions and billions of variations.

Human biology is actually far more complicated than we imagine. Everybody talks about the genes that they received from their mother and father, for this trait or the other. But in reality, those genes have very little impact on life outcomes. Our biology is way too complicated for that and deals with hundreds of thousands of independent factors.

HEMANT CHAWLA



**DRIVEN BY DNA:** Venter with chairperson Samir Brahmachari, chief of Council of Scientific and Industrial Research

Genes are absolutely not our fate. They can give us useful information about the increased risk of a disease, but in most cases they will not determine the actual cause of the disease, or the actual incidence of somebody getting it. Most biology will come from the complex interactions from all the proteins and cells working with environmental factors, not driven directly by the genetic code.

The human genetic code is being read accurately for the first time in history. What we need to do now is capture the phenotypical information—all the information that would describe each one of us physically, mentally, historically and medically. Digitising that information can help us understand the old nature versus the nurture debate. Because it can also tell us how important environmental factors are. The best way to get at the environmental component is to understand what's genetic.

Last year we published a paper on gene transplant—transplanting a genome from one organism to another and completely changing one species into another. Our researchers purified a chromosome from one bacterial species, eliminated by digestion all the proteins associated with it. We transplanted that chromosome into another bacterial species, and then selected for cells with the new chromosome. What happened is, the original chromosome got digested by enzymes and the new chromosome took over the cell. All the characteristics of one species completely went away and the species got transformed into the new one.